Bahir Dar University

Bahir Dar Institute of Technology

Faculty of Electrical and Computer Engineering

Communication and Electronics Engineering Stream

Title: Multi-user MIMO-OFDMA Channel Capacity Enhancement with constant bandwidth

Name	**ID.No**
1. Tsegaye Geta	BDU0801375UR
2. Temesgen Wale	BDU0801312UR

Name of Mentor: Mr. Tadie Birhan Medimem

January, 2020

Bahir Dar, Ethiopia

Declaration

We are students of Bahir Dar University in Bahir Dar Institute of technology (BIT), faculty of Electrical and Computer Engineering. The information found in this proposal project is our original work. And all sources of materials that will be used for the project work will be fully acknowledged.

	Name	signature
1.	Tsegaye Geta	____________
2.	Temesgen Wale	____________

Date of submission: 17/05/2013 E.C

This thesis proposal has been submitted for examination with our approval as a university advisor.

Project advisor	Signature	Date
Mr. Tadie Birhan Medimem	_______________	_________________

Acknowledgment

First and foremost, we would like to express our special sincere thanks and praise to our Almighty God and saint Merry for helping us in each day, in each time, in every minute, and in every second until we accomplish our final project.

Second, we would like to express our sincere gratitude to our advisor Mrs. Hiwot B. for her invaluable guidance and steadfast support during the course of this project. Fruitful and rewarding discussions with her on numerous occasions have made this work possible. It has been a great pleasure for us to work under her guidance. We would also like to express sincere thanks to all the faculty members of Electrical Engineering Department for their kind co-operation. We would like to acknowledge the assistance of all our friends in the process of completing this work. Finally, we express our sincere gratitude to our friends for their constant encouragement and support.

Abstract

Multiple-input multiple-output orthogonal frequency-division multiplexing access is the dominant air interface for 4G and 5G broadband wireless communications. It combines multiple-input multiple-output technology, which multiplies capacity by transmitting different signals over multiple antennas, and orthogonal frequency-division multiplexing, which divides a radio channel into a large number of closely spaced sub channels to provide more reliable communications at high speeds. Mainly multiple-input multiple-output orthogonal frequency-division multiplexing is the foundation for most advanced wireless local area network and mobile broadband network standards because it achieves the greatest spectral efficiency, and therefore delivers the highest capacity and data throughput.

The bandwidth consumption and the transmitting power are the scarce resources we need to conserve to improve the capacity and data throughput; then by keeping these parameters constant, we are going to use some techniques associated with multiple-input multiple-output concept and the orthogonal frequency-division multiplexing access concepts such as the beam forming technology to enhance the SNR with constant transmitting power by eliminating the interfering signal.

To do this project, we first start from identifying the problems on the basis of data rates and throughput especially on areas with group of people such as stadiums and restaurants. Then we will try to execute the multiple-input multiple-output orthogonal frequency-division multiplexing with different schemes including the multiple-input multiple-output, orthogonal frequency-division multiplexing and both multiple-input multiple-output orthogonal frequency-division multiplexing.

Multiple-input multiple-output orthogonal frequency-division multiplexing is expected to have high channel capacity and data throughput when compared to the multiple-input multiple-output and orthogonal frequency-division multiplexing separately. This is because orthogonal frequency-division multiplexing technique has high data rate and the multiple-input multiple-output makes the system to have more users.

Table of Contents

Acronyms

AP… .... Access Point

BER…..................................... Bit Error Rate

CSI .... Channel State Information

HD.. High Definition

HEW .... High-Efficiency Wireless

ICI .... Inter-Carrier interference

LAN… .... Local Area Network

LTE… .... Long Term Evolution

MCS .... Modulation and Coding Set

MU-MIMO… .... Multi User Multi Input Multi Output

OFDMA .... Orthogonal Frequency Division Multiple Access

QAM… .... Quadrature Amplitude Modulation

RF .... Radio Frequency

RU….................................... Resource Unit

Rx.. Receiver

STC… .... Space Time Coding

STFC .... Space Time Frequency Coding

SU-MIMO….............................. Single User Multi Input Multi Output

TX .... Transmitter

WLAN .... Wireless Local Area Network

List of figures

List of Tables

Chapter One

1. Introduction

1.1 Background of the Project

Recent advances in mobile computing and hardware technology enable transmission of rich multimedia contents over wireless networks. Examples include digital TV, voice and video transmission over cellular and wireless LAN networks, and sensor networks. With the high demand for such services, it become crucial to identify the system limitations, define the appropriate performance metrics, and to design wireless systems that are capable of achieving the best performance by overcoming the challenges posed by the system requirements and the wireless environment. In general, multimedia wireless communication requires transmitting analog sources over fading channels while satisfying the end-to-end average distortion and delay requirements of the application within the power limitations of the mobile terminal [1],[2].

Multiple input multiple output wireless communication system uses the multiple antennas at both sender and receiver side to improve communication performance. Multiple antennas at the transmitter and receiver introduces signaling degrees of freedom that were absent in SISO systems. It is the one of the smart antenna technologies available in several forms. Orthogonal frequency division multiplexing access is another popular radio technology for providing the solution for spectrum challenges. But these technologies individually will not give the higher information rates. With OFDMA, a single channel within a spectrum band can be divided into multiple, smaller sub-signals that transmit information simultaneously without interference. Because MIMO technology is able to link together many smaller antennas to work as one, it can receive and send these OFDM multiple sub-signals in a way that allows the bandwidth to be substantially increased to each user as required [3].

The bandwidth of wireless communication systems is often limited by the cost of the radio spectrum required. Any increase in bit rate, which can be realized without increasing the bandwidth, makes the system more spectrally efficient and less costly. Previous wireless communication systems have been made more spectrally efficient through the use of clever coding techniques and algorithms. However, the fundamental bandwidth limitation does not change. MIMO OFDMA communication systems have the ability to greatly increase spectral efficiencies. As opposed to previous wireless systems, in which there is one transmitting and one receiving antenna (SISO), MIMO systems use arrays of multiple antennas at both ends of the communication link, all operating at the same frequency at the same time. This introduces spatial diversity into the system, which can be used to tackle the problem of multipath.

In wireless communications system, such as point to point radio links, radio waves do not simply propagate from the transmit antenna to the receive antenna. Rather they bounce and scatter off objects, this effect is known as multipath. This effect is regarded as an impediment to the accurate transmission of data in previous wireless links. MIMO systems exploit multipath by using the rich scattering environment to increase the spectral efficiency of the wireless system. In wireless scenario, the radio channel bandwidth is an inadequate resource and that is why the availability of faster transmission and reception is restricted. In this era people wants to have accurate communication of their messages with ultimate speed but with a sufficient low cost. For making the current system efficient the degradation in service quality can no more be the excuse. Right now the race is towards achieving the goal of excellent speed of propagation with lowest BER and data rate communication with lowest error rate and of course the bandwidth requirement must be reasonable enough. But because of certain practical limitations of wireless channel i.e. quoted as multipath structure, fading, effect of Doppler shift, inter symbol interference, etc., the efficiency of the existing standards cannot be upgraded. Hence the above mentioned loop holes of existing technologies leads to the development of modern efficient wireless systems such as Wi-Fi, WI-MAX, LTE, which fits as the futuristic system with highest data rates along with best qualities at lowest error rate. To satisfy the requirement of the highest capacity along with excellent error rates in the modern wireless systems, ordinary transceiver scheme i.e. single antenna at the transmitter and receiver side is not at all justifiable [3].

The implementation of various advance antenna systems by suitable approaches is the most innovative aspect to satisfy the above discussed needs. MIMO technology means multiple antennas at both ends of a communication system [3], [4]. The idea behind MIMO is that the transmitter antennas at one end and the receiver antennas at the other end are connected and combined in such a way that the bit error rate, SNR or the data rate for each user is improved. MIMO has the capacity of producing independent parallel channels and transmitting multipath data streams and thus meets the demand for high data rate wireless transmission. This system can provide high frequency spectral efficiency and is a promising approach with tremendous potential. The use of multiple antennas at both transmitter and receiver has been shown to be an effective way to improve capacity and reliability over those achievable with single antenna wireless systems [5].

This project mainly studies software implementation of MU MIMO OFDMA based channel capacity enhancement with constant band width and power.

1.2 Statement of Problem

The wireless communication environment is very hostile. The signal transmitted over a wireless communication link is susceptible to fading (severe fluctuations in signal level), co-channel interference, and dispersion effects in time and frequency, path loss effect, etc. On top of these problems, the limited availability of bandwidth possess a significant challenge to a designer in designing a system that provides higher spectral efficiency and higher quality of link availability at low cost.

Previous wireless communication systems such as, SISO, SIMO, MISO and even the SU MIMO have been made more spectrally efficient through the use of clever coding techniques and algorithms. While using SISO, SIMO, MISO and even the SU MIMO there are some limitations in case of the data rates, throughput and the use of bandwidth in wireless communications and mobile communications especially in Wi-Fi and wireless broadband (WI-MAX), then in this case we are going to use MU MIMO OFDMA concept to improve the channel capacity with constant bandwidth and transmitting power due to the scarcity of these resources.

1.3 Objective of the project

1.3.1 General Objective

To enhance the channel capacity of MU MIMO OFDMA based wireless communication system using mathematical equation and MATLAB software with constant bandwidth.

1.3.2 Specific Objective

- To model the MU MIMO OFDMA system.
- To simulate the channel capacity versus SNR with different number of antenna, channel capacity versus SNR for MIMO OFDMA and MIMO OFDMA system, effect of beam forming on signal frequency spectrum, and bit error rate.
- To analyze the simulation results.

1.4 Scope of the Project

We execute the simulation with MATLAB to display channel capacity verses SNR with different number of antennas, channel capacity versus SNR for MIMO, OFDMA and MIMO-OFDMA system, the effect of beam forming and non-beam forming on BER, equalize the original transmitted signal frequency spectrum with the estimated signal frequency spectrum by using beam forming technique.

1.5 Significance of the Project

MU MIMO OFDMA based wireless communication has numerous applications in real life. Some of them are;

- Increases the communication reliability
- Increase the data rate
- It is applicable for highly popular areas
- Bandwidth gain
- Power efficiency

1.6 Project Organization

This section will give an outlines of the structure of the project. This project will consist of five chapters including this chapter. The following is an explanation for each chapter. Chapter 2 discuss about literature review. Chapter 3 discusses the basic concept of MU MIMO OFDMA based wireless communication system in term of definition, algorithm, system modeling and mathematical algorithm for the system proposed. Chapter 4 consists of MATLAB software results and results analysis. Comparison between each graphically result was done. The last chapter (5) summarizes the overall conclusion for this project and a few suggestion and recommendation for future development.

Chapter Two

2. Literature Review

MU MIMO OFDMA transmission system have been comprehensively studied over the past decade by numerous researchers, we have been proposed to increasing demand for high data rates, high channel capacity and increase spectral efficiencies.

We study the MIMO broadcast channel and compare the achievable throughput for the optimal strategy of paper coding to that achieved with sub-optimal and lower complexity linear pre-coding (e.g., zero-forcing and block diagonalization) transmission [6]. Both strategies utilize all available spatial dimensions and therefore have the same multiplexing gain, but an absolute difference in terms of throughput does exist. The sum rate difference between the two strategies is analytically computed at asymptotically high SNR, and it is seen that this asymptotic statistic provides an accurate characterization at even moderate SNR levels. Weighted sum rate maximization is also considered, and a similar quantification of the throughput difference between the two strategies is computed. In the process, it is shown that allocating user powers in direct proportion to user weights asymptotically maximizes weighted sum rate.

In [4], this paper presents two such constrained solutions. The first, referred to as "block diagonalization," is a generalization of channel inversion when there are multiple antennas at each receiver. It is easily adapted to optimize for either maximum transmission rate or minimum power, and approaches the optimal solution at high SNR. The second, "successive optimization," is an alternative method for solving the power minimization problem one user at a time, and it yields superior results in some (e.g., low SNR) situations. Both of these algorithms are limited to cases where the transmitter has more antennas than all receive antennas combined. In order to accommodate more general scenarios, we also propose a framework for coordinated transmitter-receiver processing that generalizes the two algorithms to cases involving more receive than transmit antennas.

In [10], this paper proposes the network spectral efficiency in terms of bits/s/Hz of a MIMO adhoc network with K simultaneous communicating transmitter-receiver pairs. Assume that each transmitter is equipped with 't' antennas and each receiver with 'r' antennas and each receiver implement single user detection. With CSI corresponding to the desired receiver available at the transmitter, it demonstrates the asymptotic spectral efficiency. Asymptotically optimum signaling is also derived under the same CSI assumption, i.e., each transmitter knows the channel corresponding to its desired receiver only. Further capacity improvement is possible with stronger CSI assumption; we demonstrate that, this using a heuristic interference suppression transmit beam forming approach.

In [6] the performance of space time frequency coded spatial modulation system is investigated with the use of code division multiple access using a simulation on MATLAB program of bit error rate. The system under study uses four antennas at both side transmitter and receiver to deploy a multipath system, where the channels in each receiver is considered to be uncorrelated channel, which mean any change in each channel doesn't affect the other channels.

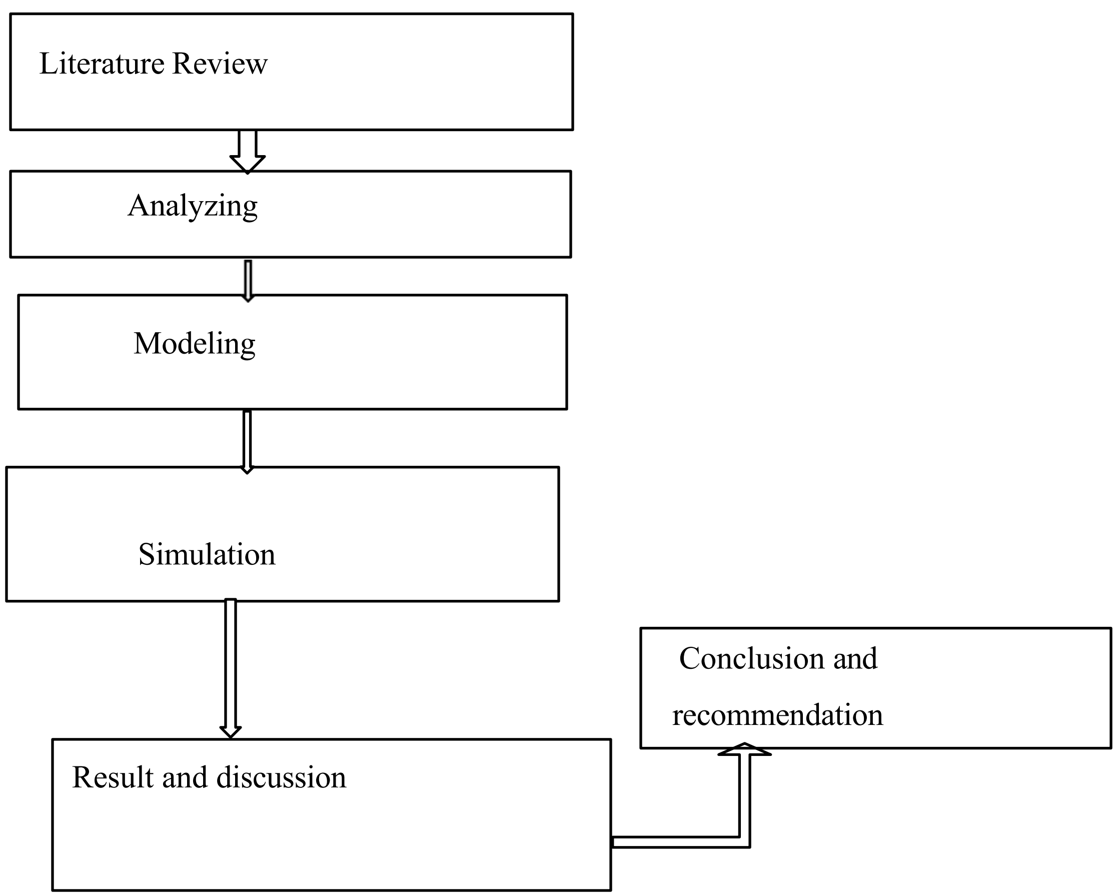
As the collected literatures indicate all the researchers did their investigation by using single approach to minimize specific effect. But in this project, MU MIMO OFDMA based channel capacity enhancement with constant bandwidth and transmitting power performed by using MATLAB with beam forming and precoding approach to have more users with the desired data rate.

Chapter Three

3. System Methodology and Mathematical Modeling

3.1 Methodology

The methodology used in doing this project would have comprised important procedures to achieve its objectives. The procedures are developed by reviewing and studying of various literatures on performance and capacity enhancement of MU MIMO OFDMA wireless communication system. These procedures are analysis, modeling, simulation, Result and discussion, and Conclusion and recommendation. The procedures involved simulation of the substrate parameter using MATLAB Software. By using multiple antennas and precoding the data, different data streams could be sent over different paths. Rayleigh suggested and later proved that the processing required by MIMO at higher speeds would be most manageable using OFDM modulation, because OFDA converts high speed data channel into a number of parallel lower speed channels.



3.2 Expected Output

- We will see the effect of path loss in throughput when we do in only MU MIMO, only OFDM and in MU MIMO-OFDM.
- We will see the effect when the number of modulation order and the number of transmitting antennas varying on channel capacity.
- We will see the effect of signal with and without beam forming on BER.

3.3 Channel Capacity

The channel capacity, C, is defined to be the maximum rate at which information can be transmitted through a channel. According to the channel capacity equation,

$C = B^* \log_2 (1 + S/N)$ bit/s...(2.1)

$S/N = n^2 * \text{Signal power}/n. \text{(noise)} = n * SNR$… ....(2.2)

So the channel capacity

$C = B * \log_2 (1 + n * SNR)$ bit/s…..(2.3)

$n = Nr * Nt$

Where, C-capacity, B-bandwidth of the channel, S-signal power, N-noise power, n-number of antenna, Nr-number of receiver antenna, Nt- number of transmitting antenna

When B tends to infinity, capacity saturates to 1.44S/N. What happens when S/N become infinity according to the formula above, capacity should become infinite, meaning a finite bandwidth supports infinite data rates at very large SNR. On the other hand, if we take Inter Symbol Interference into account, capacity should saturate to 2B at very large S/N.

In comparison with standard single antenna system, the multiple antenna systems channel capacity with Nt transmit and Nr receive antennas can be uplifted by a multiple of minimum value of (Nt, Nr) without the usage of extra transmit power and increased bandwidth spectrum. Since faster data transmission is on high demand in the near future telecommunication systems, the investigation of multiple antenna systems is actively going on.

802.11ax WLAN

802.11ax WLAN is the first WLAN standard to use OFDMA to enable transmissions with multiple users simultaneously (it is called high efficiency multi users access*)*. In OFDMA, a symbol is constructed of subcarriers where the total number defines a protocol data unit bandwidth. Each user is assigned different subsets of subcarriers to achieve simultaneous data transmission in MU environment.

802.11ax has the challenging goal of improving the average throughput per user by a factor of at least four times in dense user environments. This new standard focuses on implementing mechanisms to serve more users a consistent and reliable stream of data (average throughput) in the presence of many other users.

High-efficiency wireless includes the following key features;

- Increase four times the average throughput per user in high-density scenarios, such as train stations, airports and stadiums.
- Data rates and channel widths similar to 802.11ac, with the exception of new Modulation and Coding Sets with 1024-QAM.
- Improved traffic flow and channel access
- Better power management for longer battery life

The following table highlights the most important changes to this revision of the standard, in contrast to the current 802.11ac implementation:

Table 3. 1 802.11ac vs.802.11ax

	802.11ac	802.11ax
Bands	5GHZ	2.4GHZ AND 5GHZ
channel band width	20MHZ,40MHZ,80MHZ,160MHZ	20MHZ,40MHZ,80MHZ 160MHZ
FFT size	64,128,256,512	256,512,1024,2048
subscriber spacing	112.5KHZ	78.125KHZ
OFDM Symbol duration	3.2US+0.8/0.4USCP	12.8US+0.8/1.6/3.2USCP
highest modulation	256QAM	1024QAM
data rates	433Mbps(80MHZ,1SS) 6933Mbps(160MHZ,8SS)	600.4Mbps(80MHZ,1SS) 9607.8Mbps(160MHZ,8SS)

The 802.11ax specification introduces significant changes to the physical layer of the standard. However, it maintains backward compatibility with 802.11a/b/g/n and /ac devices, such that an 802.11ax subscriber access terminals can send and receive data to legacy subscriber access terminals. These legacy clients will also be able to demodulate and decode 802.11ax packet headers though not whole 802.11ax packets and back off when an 802.11ax subscriber access terminal is transmitting.

Multi user operation

The 802.11ax standard has two modes of operation:

- **Single User**: in this sequential mode, the wireless subscriber access terminal send and receive data one at a time once they secure access to the medium.
- **Multi-User**: this mode allows for simultaneous operation of multiple non-AP subscriber access terminals. The standard divides this mode further into downlink and uplink multi-user.

Downlink multi-user refers to the data that the AP serves to multiple associated wireless subscriber access terminals at the same time. The existing 802.11ac standard already specifies this feature. And uplink multi-user involves simultaneous transmission of data from multiple subscriber access terminals to the AP. This is new functionality of the 802.11ax standard, which did not exist in any of the previous versions of the Wi-Fi standard. Under the multi user mode of operation, the standard also specifies two different ways of multiplexing more users within a certain area: OFDMA. For both of these methods, the AP acts as the central controller of all aspects of multi-user operation. An 802.11ax AP can also combine MU-MIMO with OFDMA operation.

3.4 MIMO in Wireless Communications System

It uses the multiple antennas at both sender and receiver side to improve communication performance. It is the one of the smart antenna technologies available in several forms. The MIMO technology is the one of interested technology in wireless communications, because it significantly increases the data throughput and link range too without the additional bandwidth or increased transmit power. Multiple antennas at the transmitter and receiver introduces signaling degrees of freedom that were absent in SISO systems. This is referred to as the spatial degree of freedom. The spatial degrees of freedom can either be exploited for “diversity” or “multiplexing” or a combination of the two. In simple terms, diversity means redundancy.

A simple example of diversity is multiple antennas trying to receive the same signal,[7],[8]. The received signal on the two antennas is corrupted by noise that is uncorrelated between antennas; therefore, by combining the two signals a better quality signal can be reconstructed [9]. The analogy here is that by looking at the same object from two different vantage points, richer information on the object can be obtained. Diversity can also be achieved by using multiple transmit antennas by using space time coding techniques, when a MIMO transmitter/receiver pair operates in an environment rich in scattering, the channel matrix becomes invertible, thus enabling the receiver to decode all the different signal transmitted from the various transmit antenna apertures.

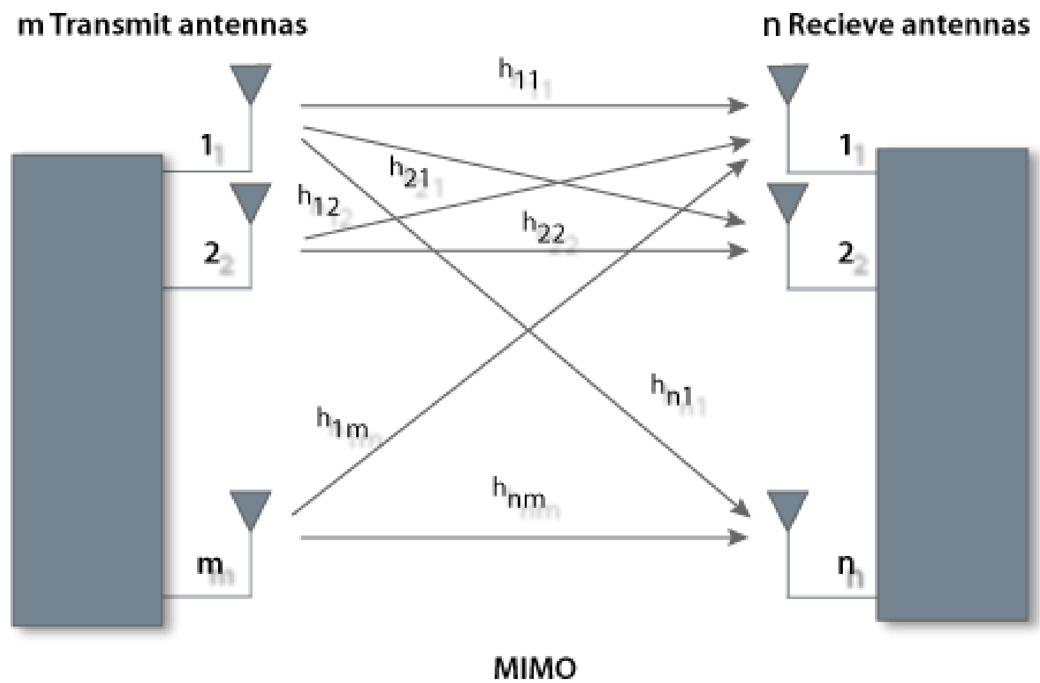


Figure 3. 1 Block diagram of MIMO

SU MIMO

The most obvious down side to single user MIMO is that the multiple streams of data must be sent or received between just one devices at a time. However, there are more advantages as well.

Figure 3. 2 SU MIMO

For instance, single user MIMO requires both the transmitting and receiving Wi-Fi radios support the MIMO technology, along with having the multiple antennas. The multiple antennas add cost, weight, and size to the Wi-Fi devices and the processing of the MIMO signals requires more resources as well. These are especially evident with the smaller devices, like smart phones and tablets.

MU MIMO

Multiple streams of data can be sent or received to or by different devices.

Figure 3. 3 MU-MIMO

Borrowing from the 802.11ac implementation, 802.11ax devices will use beam forming techniques to direct packets simultaneously to spatially diverse users.

3.4 OFDMA in Wireless Communications System

The 802.11ax standard borrows a technological improvement from 4G cellular technology to multiplex more users in the same channel bandwidth: i.e. OFDMA. Building on the existing OFDM digital modulation scheme that 802.11ac already uses, the 802.11ax standard further assigns specific sets of subcarriers to individual users. That is, it divides the existing 802.11 channels (20, 40, 80 and 160 MHz wide) into smaller sub-channels with a predefined number of subcarriers. Also borrowing from modern long-term evolution terminology, the 802.11ax standard calls the smallest sub channel a RU with a minimum size of 26 subcarriers.

Based on multi-user traffic needs, the AP decides how to allocate the channel, always assigning all available RUs on the downlink [6]. It may allocate the whole channel to only one user at a time just as 802.11ac currently does or it may partition it to serve multiple users simultaneously,[10],[11].

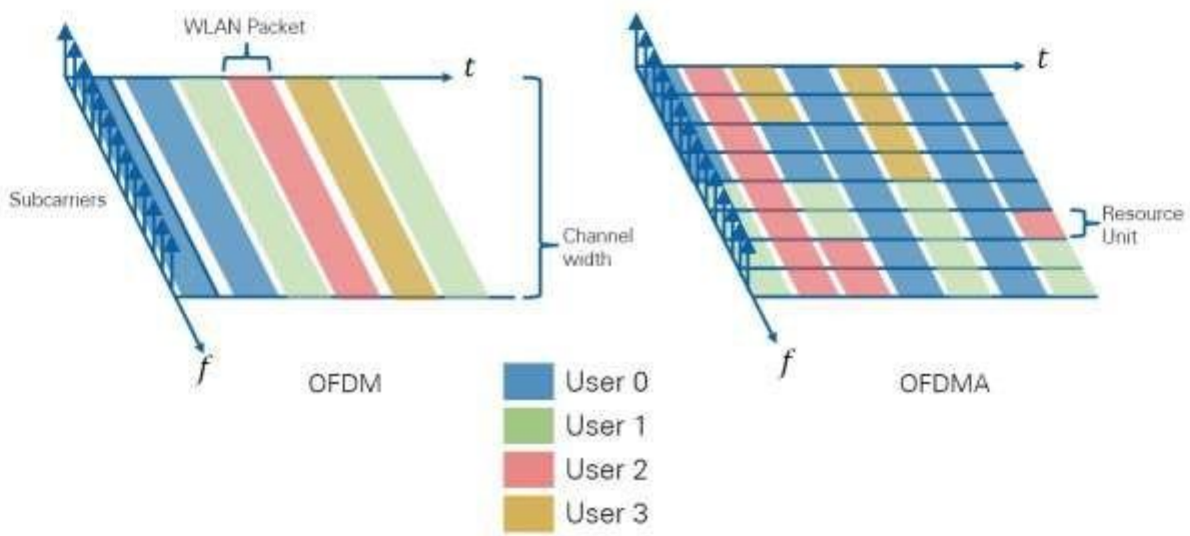


Figure 3. 4 A single user using the channel vs. multiplexing various users in the same channel using OFDMA

In dense user environments where many users would normally contend inefficiently for their turn to use the channel, this OFDMA mechanism now serves them simultaneously with a smaller but dedicated sub channel, thus improving the average throughput per user. Note that the smallest division of the channel accommodates up to nine users for every 20MHz of bandwidth.

In OFDMA, different transmit powers may be applied to different RUs. There are maximum of 9 RUs for 20 MHz, 18 RUs in case of 40 MHz and more in case of 80 or 160 MHz. The RUs enables an access point station to allow multi-users to access it simultaneously and efficiently.

There are three subcarrier types used in OFDMA WLAN;

a) **Data subcarrier**: used for actual data transmission.

b) **Pilot subcarrier**: used for phase information and parameter tracking.

c) **Unused subcarrier**: which is neither data nor pilot subcarrier this includes Guard band and null subcarriers.

- **Data mapping:** is the process of creating data element mappings between two distinct data models. Data mapping is used as a first step for a wide variety of data integration tasks, including;
- Data transformation or data mediation between a data source and a destination
- Identification of data relationships as part of data lineage analysis
- Discovery of hidden sensitive data such as the last four digits of a social security number hidden in another user id as part of a data masking or de-identification project.
- Consolidation of multiple databases into a single database and identifying redundant columns of data for consolidation or elimination.

The term cyclic prefix refers to the prefixing of a symbol, with a repetition of the end. The receiver is typically configured to discard the cyclic prefix samples, but the cyclic prefix serves two purposes;

- It provides a guard interval to eliminate inter symbol interference from the previous symbol.
- It repeats the end of the symbol so the linear convolution of a frequency-selective

The multipath channel can be modeled as circular convolution, which in turn may transform to the frequency domain via a discrete fourier transform. This approach accommodates simple frequency domain processing, such as channel estimation and equalization.

For the cyclic prefix to serve its objectives, it must have a length at least equal to the length of the multipath channel. The concept of a cyclic prefix is traditionally associated with OFDM systems; however, the cyclic prefix is now also used in single carrier systems to improve the robustness to multipath propagation.

3.5 Mu MIMO-OFDMA in Wireless Communications System

MIMO is an advanced antenna strategy for next generation remote access that can carry number of times more data traffic than today's most advanced third generation networks. OFDMA is another popular radio technology for providing the solution for spectrum challenges. But these technologies individually will not give the higher information rates. With OFDMA, a single channel within a spectrum band can be divided into multiple, smaller sub-signals that transmit information simultaneously without interference. Because MIMO technology is able to link together many smaller antennas to work as one, it can receive and send these OFDM's multiple sub-signals in a way that allows the bandwidth to be substantially increased to each user as required.

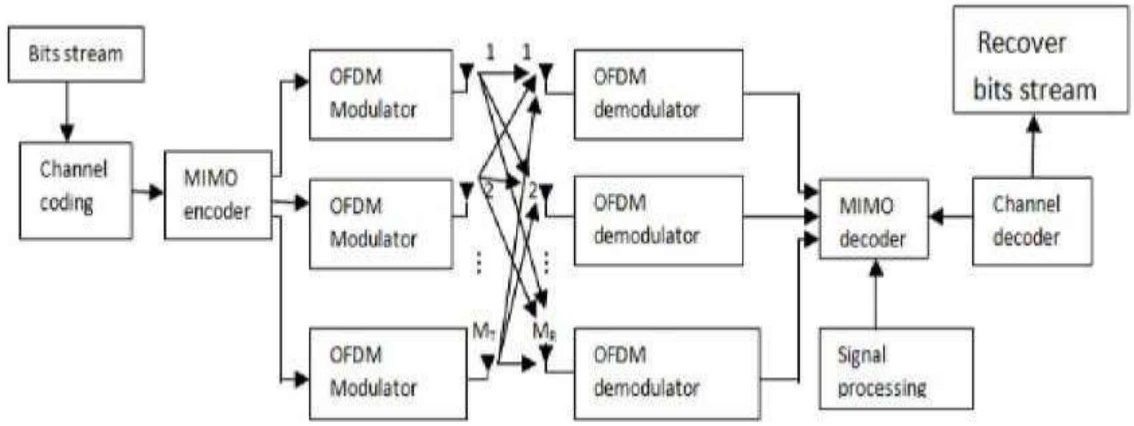


Figure 3. 5 Architecture of OFDMA with MIMO

MIMO innovation is prevalently utilized as a part of broadband framework. Bit error rate is the biggest issue in the field of communication. Due to this new innovation this problem is resolved by combining both the approaches i.e. MIMO with OFDMA.

Advantages of MIMO OFDMA;

- The major advantages are increase in channel capacity.
- The coding over the frequency, time and space area empowers on a great deal, more solid and powerful transmission over the harsh remote environment.
- The systems support high information rate and superior enhancement in diversity gain and upgraded capacity of framework.
- Power efficiency optimization,
- Enhancement in reliability and enhancement in range of covering area.

3.6 Beam Forming

The formation of beam in the desired direction with appropriate power is known as beam forming. Beam forming employs facility to vary power of the beams and their directions with the help of amplitude/phase variations. Antenna arrays are used instead of single antenna in beam forming system of MIMO OFDMA systems. Earlier days, mechanical rotation-based mechanism was employed for steering the antenna or antenna array and separate mechanism for power amplification of beams (i.e. antenna patterns) was used [11],[12],[13].

The procedures that we used are the AP will calculate a channel matrix for each user and steer simultaneous beams to different users, each beam containing specific packets for its target user. 802.11ax supports sending up to eight multi-user MIMO transmissions at a time, up from four for 802.11ac. Also, each MU MIMO transmission may have its own modulation and coding set and a different number of spatial streams. By the way of analogy, when using MU MIMO spatial multiplexing, the AP could be compared to an Ethernet switch that reduces the collision domain from a large computer network to a single port.

As a new feature in the MU MIMO uplink direction, the AP will initiate a simultaneous uplink transmission from each of the subscriber access terminal by means of a trigger frame. When the multiple users respond in unison with their own packets, the AP applies the channel matrix to the received beams and separates the information that each uplink beam contains.

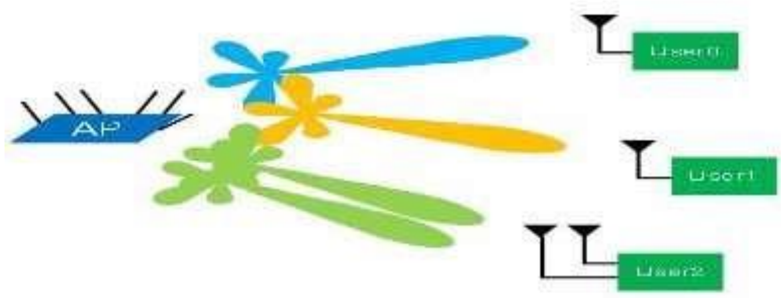


Figure 3. 6 AP using MU-MIMO beam forming to serve multiple users located in spatially diverse positions

3.7 Mathematical Modeling of the system

We are doing MU MIMO and OFDMA based channel capacity enhancement with the constant BW and transmitting power. So to do this, we would have been some models in MU MIMO and OFDMA concepts.

3.7.1 System Model

The transmitter and receiver blocks with two antennas both at the transmitter and the receiver of the MIMO OFDM system is shown in Figure (3.7). At the transmitter, the source bit streams are mapped, passed via the space-time code encoder and then mapped to their corresponding symbols with two outputs. Each symbol passes through a serial to parallel (S/P) converter and becomes parallel data. Afterwards, the inverse fast fourier transform is performed, employing OFDM modulation in each sub-carrier. Time domain symbols are parallel to serial (P/S) converted and then a cyclic prefix (CP) is added in order to mitigate or eliminate ISI and ICI. Lastly, the amplified modulated signals are transmitted.

Generally MIMO OFDMA system see below;

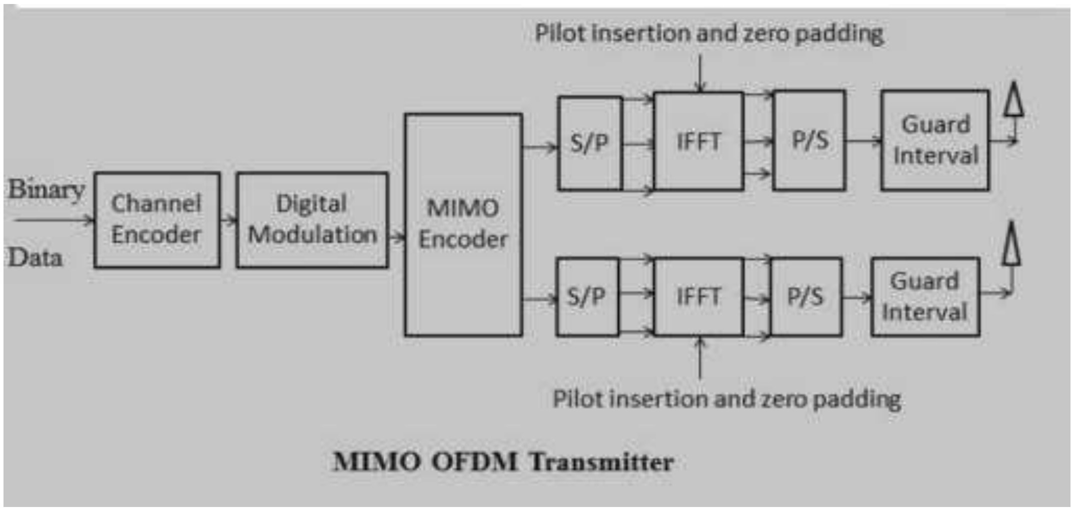


Figure 3. 7 MIMO OFDMA system transmitter with 2x2 antenna

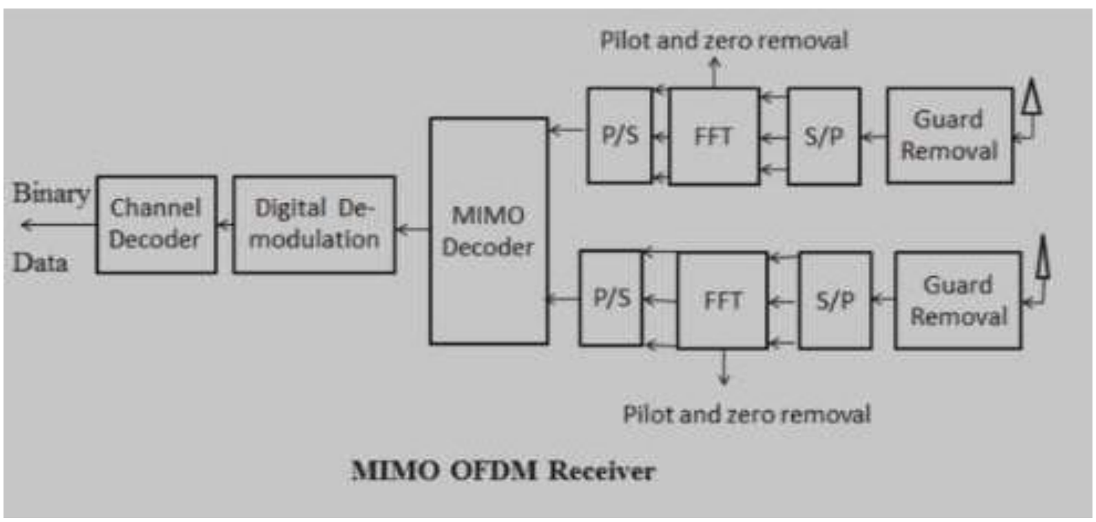


Figure 3. 8 MIMO OFDMA system receiver with 2x2 antenna

3.7.1 System Parameters

In wireless communication system we have a general formula that describes the transmitting signal, receiving signal, channel and noise signal.

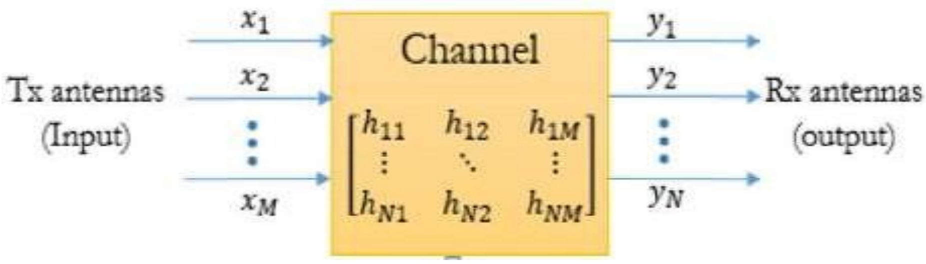


Figure 3. 9 Channel matrix

Y (t) =H (t) X (t) +n (t)...3.1

Where,

Y (t) = signal at the destination

H (t) = matrix of the channel

X (t) = signal at the input

n (t) = noise

MIMO OFDMA system with Mt transmit and Mr receive antennas, where OFDM with N subcarriers is employed at each transmit antenna. The frequency-selective fading channel between the N^{th} transmit antenna and the M^{th} receive antenna is modeled with L channel taps. Equivalently, the impulse response vector h_{NM} can be obtained by two methods these are; STF coding and beam forming methods.

3.7.2 Adaptive beam forming

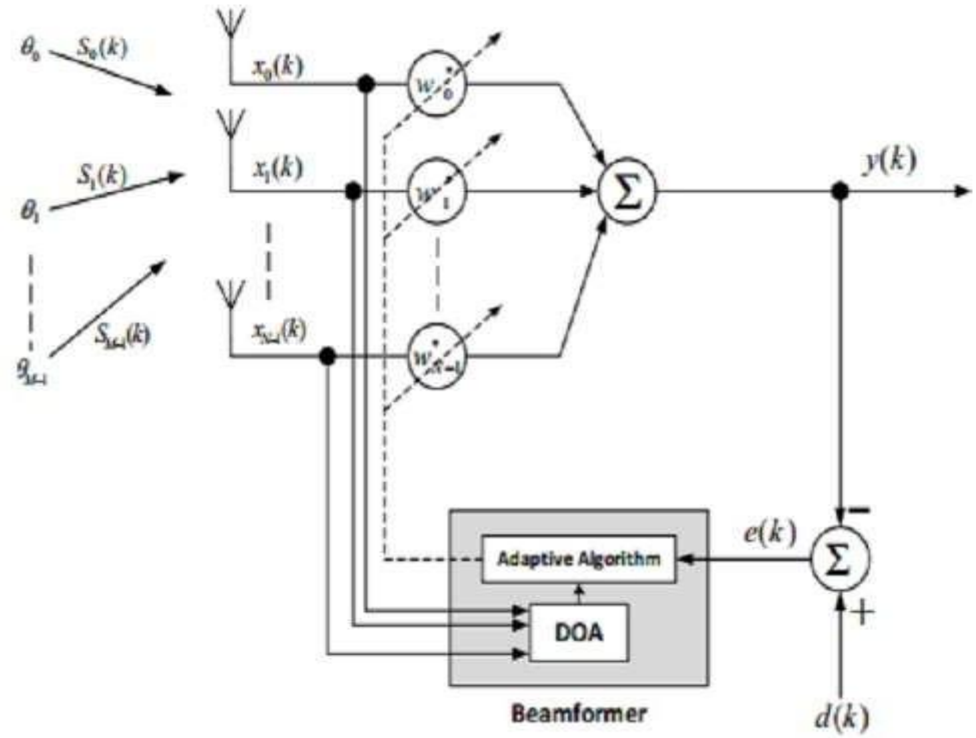


Figure 3. 10 Adaptive beam forming technique

The array factor of spherical angle θ for linear array of N radiating elements with elements spacing is given by the formulas;

$AF_{even}\ (\theta) = \sum_{n=1}^{N/2} Wncos((2n-1)u), \quad N = even$ 3.2

$AF_{\ odd}(\theta) = \sum_{n=1}^{N+1/N} Wncos(2(n-1)u), \ \ N = odd$.. 3.3

$$AF = \sum_{n=1}^{N} e^{j(n-1)\psi}$$

$$\text{where} \quad \psi = kd\cos\theta + \beta$$

Wn is the weight of N^{th} radiating element

$u = (\pi d/\lambda)\ \sin_{\theta}$..3.4

Equation (3.2) and (3.3) can be written in general form for N even or odd number of radiating elements as:

$X_n\ (t) = \sum_{I=0}^{M-1} Si(t)e^{j(-\frac{N-1}{2}+n)kdsin(\theta i)} + n_n(t)$ 3.5

Where k is wave number ($2\pi/\lambda$), λ wavelength of the incident wave, Considering M to be the number of arriving plain waves that are incident on the linear array.

From multiple directions from the incident signals on n^{th} radiating element are given by: - X_n (t) $= \sum_{I=0}^{M-1} Si(t) e^{j(-\frac{N-1}{2}+n)kd\sin(\theta i)} + N_n(t)$ 3.6

Where,

Xn (t) = the desired signals

S_i (t) = the interfering signals

Nn (t) = the received noise signal at n^{th} element.

Y (k) $= WH.X(k)$ 3.7

Where array weights vector $W = [W0 \quad W1 .... WN-1]^T$

$\bar{x}$(k): Vectors of inputs to the array which is equal to equation

$\bar{x}(k) = \bar{x}s(k) + \bar{x}i(k) + \bar{n}(k)$.. 3.8

Where, $\bar{x}(k)$= vector of desired signal

$\bar{x}i(k)$ =Vector of interfacing signal

$\bar{n}$ (k)= zero mean Gaussian noise vector

3.7.3 Adaptive Beam Forming Algorithms

The adaptive beam forming algorithms compute the weights by minimizing the error between the array output and the desired signal until the weights reach their optimum values.

In some applications, such as a wireless mobile communications system and RADAR target tracking, it is desired to achieve high speed adaptation of the weights and to reject the interference and noise.

$\hat{R}xx(k) = \alpha \hat{R}xx(k-1) + \bar{x}(k)\bar{x}^H(k)$ 3.9

$\hat{r}xx(k) = \alpha \hat{r}xx(k-1) + d^*(k)\bar{x}(k)$... 3.10

Where α is the forgetting factor; that is a positive constant value in range $0 \le \alpha < 1$. The weights of RLS algorithm are updated as follows

$\bar{W}(k) = \bar{w}(k-1) + \bar{g}(k)(d^*(k) - x^{-H}(k)\bar{W}(k-1))$ 3.11

For

$\bar{g}(k) = \alpha^{-1}\hat{R}xx^{-1}(k-1)\bar{x}(k)/1+\alpha^{-1}\bar{x}^H(k)\hat{R}xx^{-1}(k-1)\ \bar{x}(k)$ 3.12

Correlation matrix inverse can be computed iteratively as

$\hat{R}xx^{-1} = \alpha^{-1}\hat{R}^{-1}_{xx}(k-1) - \alpha^{-1}\bar{g}(k)\bar{x}^H(K)\hat{R}^{-1}_{xx}(k-1)$..3.13

The adaptive algorithm considered in this project work is based on Least Mean Square (LMS). It uses a gradient based method of steepest decent. And also estimate the gradient vector from the available data. This algorithm computes the complex weights vector recursively using the equation, given as;

$$W(n+1) = W(n) + \mu x(n)[d^*(n) - x(n)W^H] \quad 3.14$$

Where μ the step is size parameter and controls the convergence characteristics of the LMS algorithm. The LMS algorithm is initiated with an arbitrary value of W (0) for the weight vector at n=0.

The successive corrections of the weight vector eventually lead to Minimum value of the Mean Square Error (MSE). The LMS algorithm is important because of its simplicity and ease of computation, because it does not require off-line gradient estimations or repetition of data. One of the drawbacks of the LMS adaptive scheme is that the algorithm must go through many iterations before satisfactory convergence is achieved. If the signal characteristics are rapidly changing, the LMS algorithm may not allow the tracking of the desired signal in a satisfactory manner

When we are using OFDMA and MU MIMO we are finding the throughput and data rates with respect to the path loses and signal to noise ratio including different noises such as noise floor and the variation of transmitting and receiving antennas in the following formulas.

First to find out the path loss we need to have path loss formula given by

PL=A log10 (d[m]) +B+C log10 (fc [GHz]/5.0) + X… 3.15

PL free =20log10 (d) + 46.4 +20log10 (fc/5.0) 3.16

Here d is the separation between the transmitter and receiver in meters, fc is the frequency in GHz, A is the path loss exponent, B is the intercept and C is the frequency dependent parameter. X is the environment specific parameter such as path loss due to a wall. PL free is the path loss in a free space line of sight environment (here A=20, B=46.4 and C=20).

Environments can be defined the different parameters in the WINNER-II model.

Once an environment is selected the path loss parameters A, B and C can be selected from the table further down e.g. A1 is the in-building scenario with

A=18.7, B=46.8 and C=20 for the LOS case. The PL for a T-R separation of 100 m and frequency of 2 GHz is calculated as (Appendix 1):

PL=18.7*log10 (100) + 46.8+20*log10 $(\frac{2}{5})$ =76.42dB 3.17

And further from this we have used zero forcing precoding and beam forming equations and algorithms. A separate equation for the path loss is given where the parameters A, B and C are not sufficient to describe the scenario.

3.7.3 MIMO Capacity

This section, Shannon capacity results for the three algorithms under consideration will be revised. A MIMO channel is a wireless link between *MT* transmits and *NR* receive antennas. It consists of MT NR elements that represent the MIMO channel coefficients. The multiple transmit and receive antennas could belong to a single user modem or it could be distributed among different users [14].

Statistical MIMO channel models offer flexibility in selecting the channel parameters, temporal and spatial correlations. MIMO channel simulation tools are implemented based on these models. Several statistical MIMO channel models were proposed.

The matrix H is the MxN channel matrix where the element at row n and column m, hmn denotes the channel response at receiver NR due to transmitter MT. The channel capacity of MIMO system is given by:

CMIMO=B*$\log_2$[1+M.N.SNR] (bps/Hz) 3.18

We would have proposed, B is constant bandwidth which constant effect in capacity because it is limited or scares resources. Where signal to noise ratio is given by, SNR = M x N x SNR.

As per the above equations and considering M number of transmitting antennas and N number of receiving antennas, it can be said that the channel capacity of MIMO system is the highest among all diversity techniques.

3.7.4 MIMO-OFDMA Capacity

In order to determine which transmit antennas has the best SNR and channel capacity at the i^{th} iteration; the following formula for the m^{th} antenna is used:

The channel capacity using water filling solution we can determining the channel capacity of MIMO OFDMA system.

Let considering $C_D^{(i)}$ data symbol transmitted at i^{th} antenna on the D^{th} tone. Where D^{th} =(0,1,2 ....,N-1) tone

$$H(e^{j2\pi\theta}) = \sum_{i=0}^{L-1} H1\, e^{j2\pi\theta}(0 \leq \theta < 1) \text{ .... } 3.19$$

And $C_D = [\ C_D^{(0)}\ \ C_D^{(1)} \ldots\ldots\ldots C_D^{(M-1)}\ ]^T$ 3.20

is transmitted data symbol into frequency vectors it can be shown that

$$C_D^{-} = H\left(e^{j2\pi\left(\frac{k}{n}\right)}\right) CD + nD, \quad D=(0, 1,2,3\ldots\ldots\ldots n-1) \ldots\ldots 3.21$$

Where C_D^{-} is the reconstructed data vector for the D^{th} tone and n_D is additive white Gaussian noise (AWGAN) satisfying

$$\varepsilon\ \{n_D n_1^{H}\}=\sigma^2 n 1N^{\sigma}[k-1] \ldots\ldots 3.22$$

From equation 3 it can be seen that the equalization requires application of narrowband receiver for each tone D =(0, 1,2,………n-1) equation 3.21 can be rewritten as

$$C^{-}=Hc+n \ldots\ldots 3.23$$

We assume that OFDM symbol transmitted use independent realization of random channel impulse response matrix and that the channel matrix remain constant within one channel use. Using equation 5 the mutual information (b/s/Hz)of OFDM based multiplexing system under an average transmitter power constraint is given by

$$I=1/N \log_2[\det(IN,n + 1/\sigma 2H\sum H^{H})] \ldots\ldots 3.24$$

Where: T_r is the trace of the matrix A. $\sum$ with $T_{r}(\sum)<=p$ is the covariance matrix of Gaussian input vector C and p is the maximum over all transmitted power. Note that mutual information is normalized by n since n data symbol are transmitted in one OFDM symbol and that we ignored the loss in spectral efficiency due to presence of cp.

The nMxnN matrix $\sum$ is block diagonal matrix given by;

$$\sum=\mathrm{diag}\{\sum_k\}_{D=0}^{n-1}$$

Where the M×M matrices are the covariance matrices of the Gaussian vectors C_D, and determine power allocation of the transmitted antenna across OFDM tone. In OFDM spatial multiplexing system statically independent data symbol are transmitted from different antenna and different tone and total available power allocation uniformly across all space frequency sub channel.

$\sum_D= (P/(M_n))I_M(D=0,1,\ldots..n-1)$ which is easily verified to result in $T_{r}(\sum)=p$ using equation 6 we there for obtain

$$I=1/n \sum_{D=0}^{n-1} ID \ldots\ldots 3.25$$

$$I=1/n\sum_{D=0}^{n-1} \log_2[\det(IN + \acute{\rho} H(e^{j2\pi(D/n)})\ H^{H}(e^{j2\pi(D/n)})] \ldots\ldots 3.26$$

The quantity I_D is mutual information in MIMO OFDMA sub channel since $H(e^{j2\pi(D/n)})$ is random, I_D is random.

From the above the capacity of MIMO OFDMA system is given below.

$C_{MIMO\text{-}OFDMA} = 1/n\sum_{D=0}^{n-1}\log_2[\det(IN + \varrho\Lambda H H^H)]$ 3.27

Where Λ =diag $\{\lambda_i(R)\}^{N-1}_{i=0}$, H is an MXN random matrix. $R=\sum_{l=0}^{L-1} R1$ $\lambda_{i(R)}$ is denoted Eigen value of R .the capacity of MIMO-OFDMA is given above

Case 1: for large M and fixed N

$(1/M)HH^H \rightarrow I_N$

$C_{MIMO\text{-}OFDMA}=\log_2[\det(IN + \bar{Q}\Lambda)]$..3.28

$\bar{Q}=M\varrho=p/N\,\sigma^2_n$

Case2: For low SNR, large M and small $\acute{\rho}$

Capacity $MIMO\text{-}OFDMA=\log_2(\prod_{i=0}^{N-1} 1 + \bar{Q}\lambda 1(R)))$

$=\log_2[1 + \bar{Q}Tr(R)]$...3.29

Case 3: For high SNR

$C_{MIMO\text{-}OFDMA}=\sum_{l=0}^{N-1}\log_2(\det(1 + \acute{\rho}\lambda 1(R))$ 3.30

3.8 Flow Chart

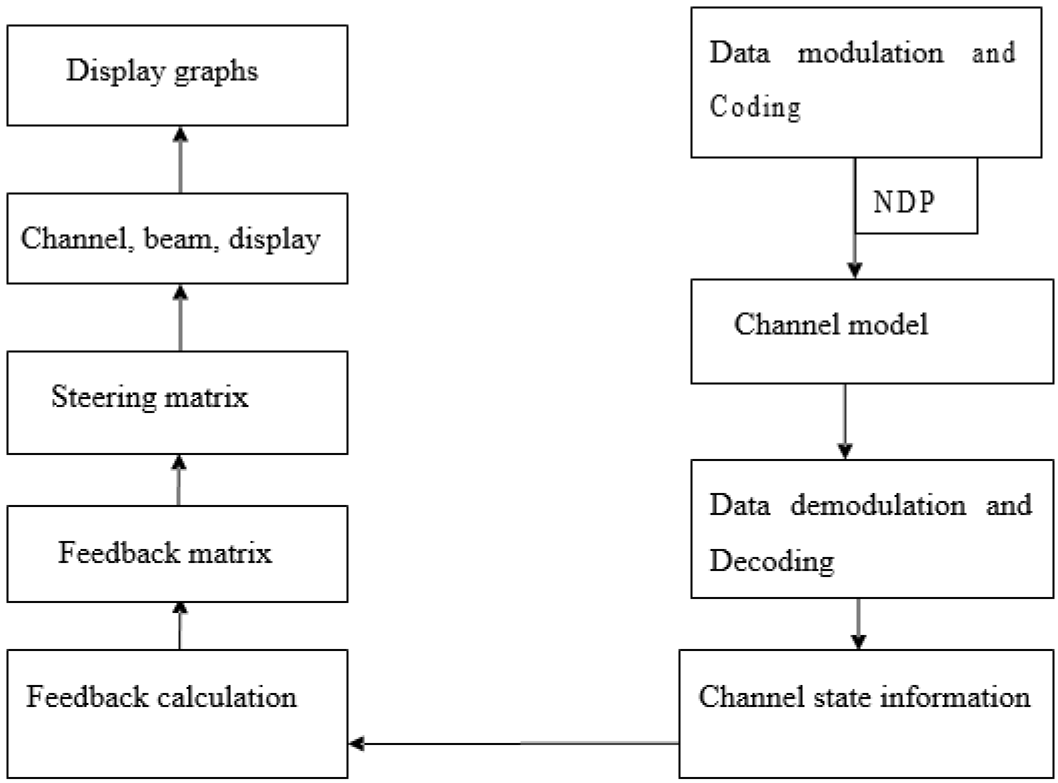


Figure 3. 11General flow chart to show how MATLAB executes the operation

Chapter Four

4. Simulation Result and Discussion

When we are doing this project, we would have used MATLAB to execute and test the parameters and the results regarding to the specified parameters. We are trying to enhance the channel capacity with the parameters of signal to noise ratio and packet error rate with constant bandwidth. The field of wireless communication systems and networks has experienced explosive growth and wireless communications has become an important part in everyday life. To overcome the limited capacity of conventional SISO, SIMO, MISO and even SU MIMO systems, by the use of multiple antennas a MIMO with OFDMA systems, offers greater capacity than SISO, SIMO, MISO and SU MIMO counterparts. The multiple antennas can be used to increase the communication reliability by diversity or to increase the data rate by spatial multiplexing or a combination of both. Multiple antennas has performance and capacity enhancements without the need for additional bandwidth or constant bandwidth.

We have used some built in functions such as in the beam forming techniques, calculating the steering matrix and the configurations of the channel model which is Winner two channel model changed to Tgax channel model and AWGN channel model. We use beam forming technique in MIMO system to reduce the effect of interfering signal. This is achieved by creating radiation pattern of the antenna array by adding the phase of signals in desired direction and by null the pattern in unwanted direction.

Table 4. 1 Simulation parameters

Parameter	Assumption
Technology	MU MIMO OFDMA
Data modulation	QAM modulation
Channel model	AWGN/Rayleigh flat fading
Transmitting bandwidth	Constant
FFT	256
Antenna configuration	2x2,3x3,4x4,5x5
SNR	The total received power per antenna to noise power ratio
Frame length	5ms
Carrier frequency	4 MHZ

Simulation Result

Figure 4.1 below shows a plot of the capacity versus SNR for MIMO OFDMA system. We note that capacity increases with increasing SNR (dB) and with increasing Nt or Nr. It is observed that at SNR=20dB the capacity varies from 6.25 bits/s/ Hz for MIMO OFDMA (NT=NR=2) to 14.25 bits/s/Hz for MIMO-OFDMA (NR=NT=5). Hence it is concluded that the capacity growth achieved by MIMO OFDMA system is the highest compared to other systems yielding remarkable improvement (especially for High SNR).

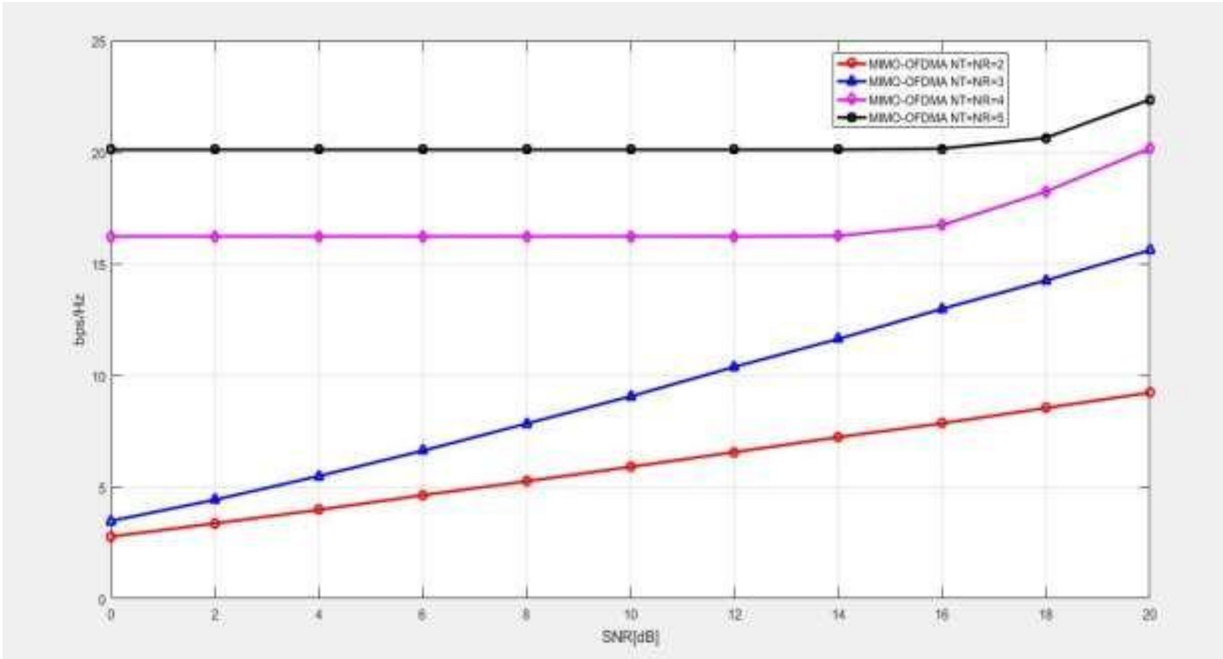


Figure 4. 1 Channel capacity verses SNR with different number of antenna

Table 4. 2 Capacity versus SNR for MIMO OFDMA System

SNR(dB)	Capacity(bit/s/HZ)			
	MIMO-OFDMA NT=NR=2	MIMO-OFDMA NT=NR=3	MIMO-OFDMA NT=NR=4	MIMO-OFDMA NT=NR=5
4	1.75	2.25	2.75	3.25
8	2.75	3.55	4.25	5.75
14	4.25	6.75	7.5	10.0
20	6.25	6.45	11.25	14.25

Figure 4.2 below shows a plot of the capacity versus SNR for MIMO, OFDMA and MIMO OFDMA System together. It is observed that at SNR=10dB the capacity varies from 2.75 bits/s/Hz for MIMO (NT=NR=2) to 7 bits/s/Hz for OFDMA (NR=NT=2) and to 14 bit/s/HZ for MIMO OFDMA (NT=NR=2).

Hence it is concluded that the capacity growth achieved by MIMO OFDMA system is the highest compared to other systems yielding remarkable improvement (especially for High SNR). Note in our project, we would have been improve channel capacity of MIMO OFDMA system at constant band width by using number of SNR. If SNR increase channel capacity is enhance as you see from simulation.

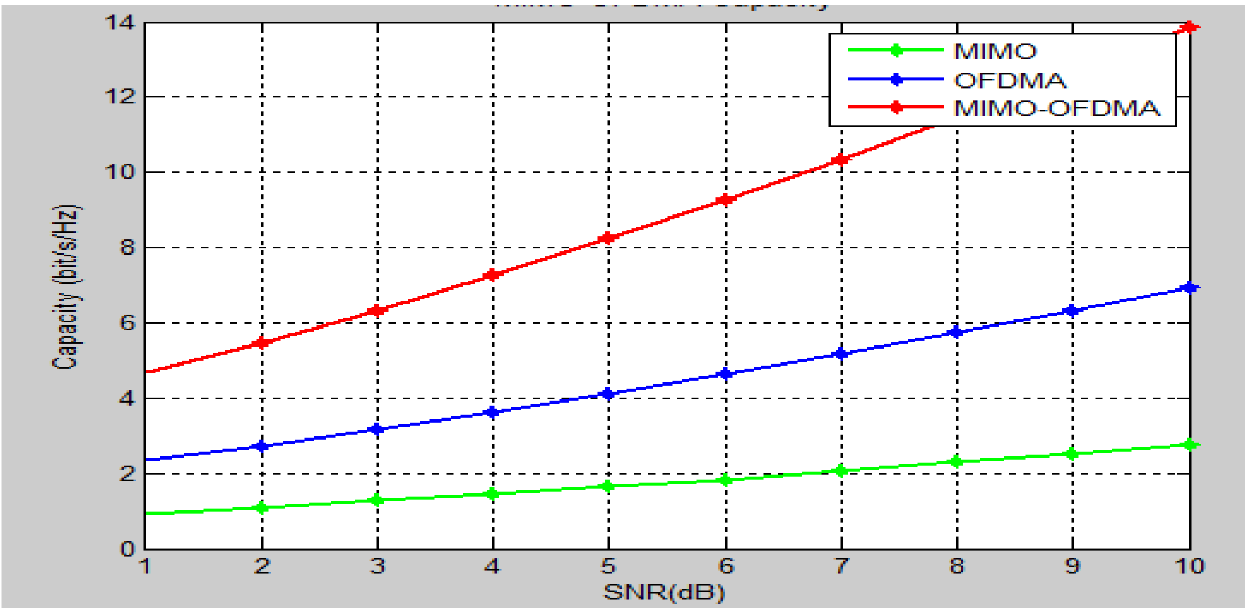


Figure 4. 2 Channel Capacity vs SNR for MIMO, OFDMA and MIMO OFDMA System

Table 4. 3 Capacity versus SNR for MIMO, OFDMA and MIMO-OFDMA System

SNR(dB)	Capacity(bit/s/HZ)		
	MIMO (NT=NR=2)	OFDMA (NT=NR=2)	MIMO-OFDMA (NT=NR=2)
4	1.75	3.75	6.55
8	2.55	5.75	11.75
10	3.25	7.25	14.0

Figure 4.3 below show the beam forming techniques the original transmitted signal frequency spectrum with the estimated signal frequency spectrum by minimizing the error between the array output and the desired signal until the weights reach their optimum values and also, by rejecting the interference.

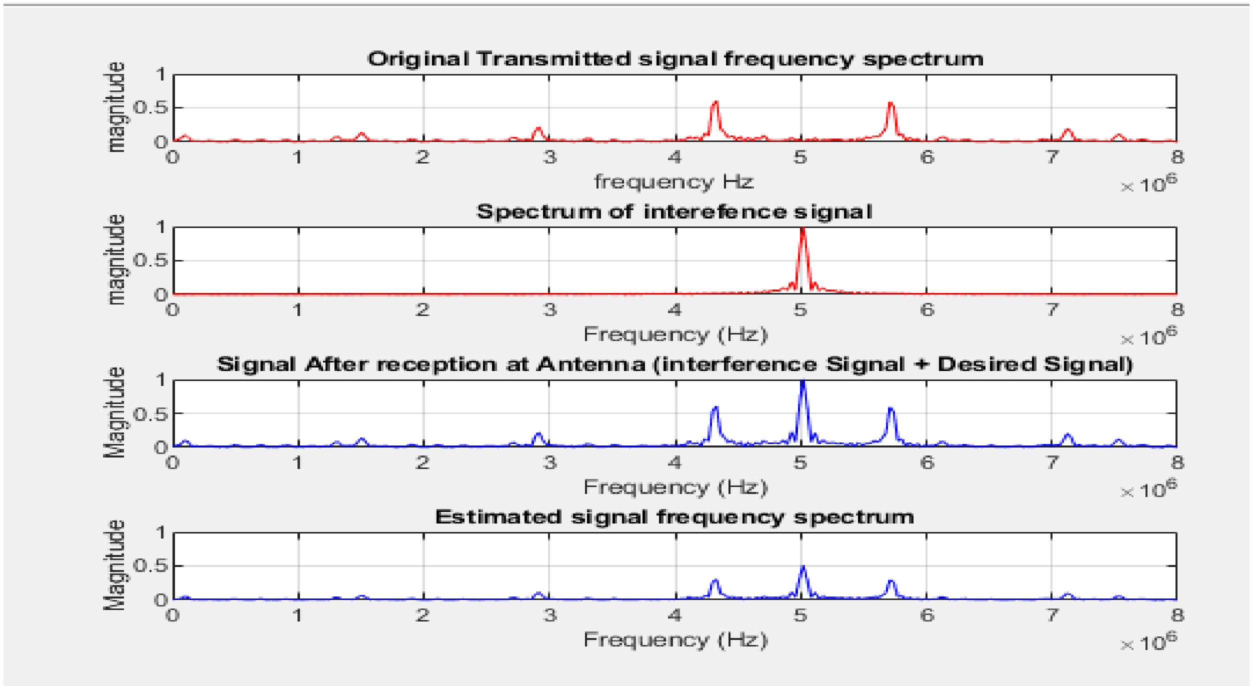


Figure 4. 3 Result of beam forming

From the above figure we have observed that beam forming technique almost equalize the original transmitted signal frequency spectrum with the estimated signal frequency spectrum by minimizing the error between the array output and the desired signal until the weights reach their optimum values and also, by rejecting the interference. We would get relatively expected signal by using beam forming spectrum.

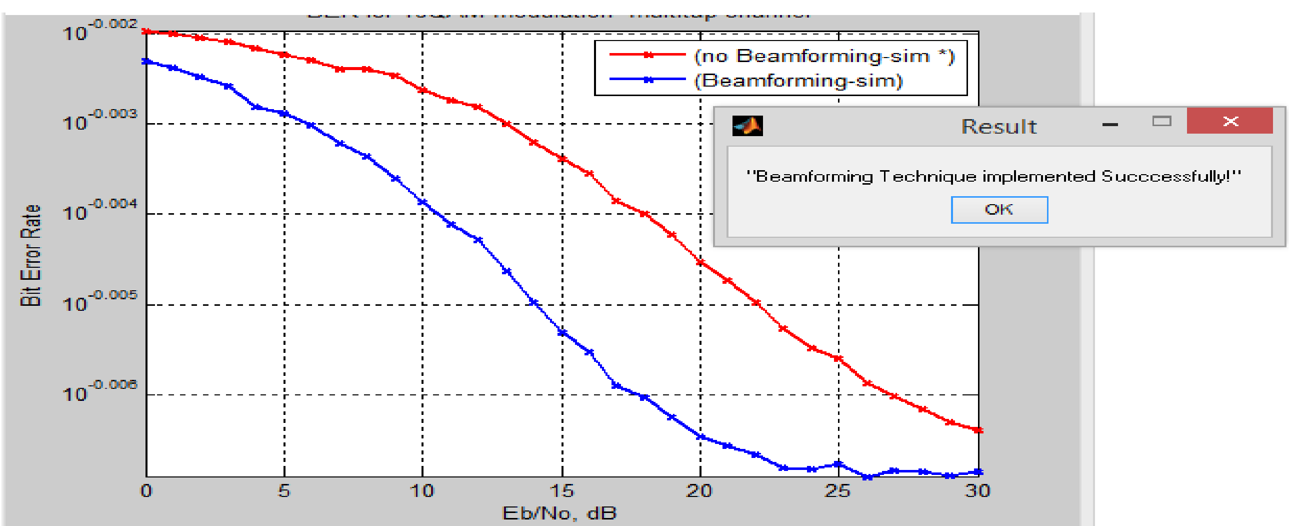


Figure 4. 4 Signal with and without beam forming

From the above Figure (4.4) we have observed that signal with and without beam forming technique by using Bit error rate and Eb/No parameter. And the signal with beam forming has low bit error rate compared to signal without beam forming and vice versa. We would concluded that if we use beam forming techniques on MIMO OFDMA System we can increases SNR by minimizing bite error rate and enhancing (improving) the capacity of the system.

Table 4. 4 BER versus SNR for 16-QAM modulation of multistep channel

SNR(dB)	BER	
	Beam forming	Non-beam forming
5	0.9938	0.9948
15	0.9884	0.9908
30	0.9875	0.9885

We can summarize from the above result and discussion in MIMO OFDMA system there is high spectral efficiency, higher SNR and low BER than MIMO only and OFDMA only.

Chapter Five

5. Conclusion and Recommendation

5.1 Conclusion

In this project we described the current situation of WLAN regarding the current channel capacity basically on the throughput and path loss faced by the access points and the devices. According to our investigation, the current WLAN technology based on SU MIMO has bandwidth limitations and this leads to have less throughput access which leads to more power consumption and time loss to the subscribers. So, we have designed a system based on MIMO OFDMA, which has the capability to enhance the channel Capacity current WLAN system with constant BW. We strongly believe that this system is much better than the previous system using MIMO only. From the simulations performed, it can be concluded that a simple way to improve the capacity of the Channel is to increase the number of receiver antennas. For small devices, where the number of antennas is limited, a solution is to increase the number of transmit chains with very little decoding complexity. When the SNR is high; diversity is important and over compensates the larger sensitivity of high-order modulation schemes.

- By considering BER as parameter that beam forming techniques is good for error less transmission.
- By using high order antenna configuration space diversity can be increased, which will further decrease the BER at given SNR as compared to lower order Antenna configurations.
- By doing so, even higher data capacity at any given SNR can be achieved.
- From the simulations performed, it can be concluded that a simple way to improve the quality of the transmission is to increase the number of receive antennas.

Generally, we concluded that when the number of antennas and the modulation order increase, and using beam forming techniques Spectral efficiency (Capacity) is enhanced for MIMOOFDMA System with constant bandwidth.

5.2 Recommendation and Future Work

In MIMO-OFDMA systems, spatial multiplexing is used where independent signals are transmitted simultaneously via different antennas. This gives good results in increasing the capacity of the channel. This well-known system is designed to maximize the spectral efficiency with constant bandwidth. However, in some environments, the independent links of MIMO system only may suffer from a considerable fading which causes decreasing in the total data rate. Space-time block codes can mitigate that problem but with occupation of at least two links to transmit one symbol in at least two time slots which decreases the number of transmitting layers and transmitted symbols. The result is a decrease in channel capacity. MIMO OFDMA systems is the key techniques for the next generation communication systems and achieves high data rates with acceptable BER performance over a good channels state.

The current project is MU MIMO OFDMA based channel capacity enhancement which describes basically the channel capacity with different number of antennas and modulation orders but still the system have some trade off especially on the SNR so we need to recommend future researchers to develop this system by combining STF coding combined with Beam forming and throughput with path loss to improve the SNR trade off and to make the MU MIMO concept in to massive MIMO then it will support up to 100 antennas.

References

[1] C.K.Kim, K. a. (2000). "Adaptive Beamforming Algorithm for OFDM Systems with Antenna Array". *IEEE Trans. on consumer Electronics*, 1052-1058.

[2] M.Lei, P. H. (May 2004). "Adaptive Beamforming based on frequency-To-Time Pilot. Transform For OFDM". *Proc. of 59th IEEE Vehicular technology conf.*, 564-656

[3] Chin, Z. a. (May 2004). "Post and Pre-FFT Beamforming in an OFDM System". *Proc.59th IEEE Vehicurar Technology Conference*, 132-234.

[4] R.Prasad, [. R. (2000). OFDM For wireless Multimedia Communication. *Artech House Pubilshers*, 354-512.

[5] Wikipedia MIMO in wireless communications, http://en.wikipedia.org/wiki/MIMO.

[6] Poongodi, P.Ramya, & Ashanmugam. (2010). "BER Analysis of MIMO-OFDM system using m-QAMover Rayligh Fading Channel. *Proceding of the international conference on communication and computational Intelligence*, 702-756.

[7] A.udwatl, p. S. (2011). Comparation of optimization capabilityes of RLS Adaptive Beamforming Algorizmfor smart antenna a system. *Pro National conference* , 78-90.

[8] Chin, Z. a. (2004). "Post and pre FFT Beamforming in an OFDM System ". *Proo IEEE Vehicular Technology conference*, 89-100.

[9] A.Forouzan, [. B. (2007). *data communications and Networking.* New York: McGraw- Hill,.

[10] R.Bouallegue, [. A. (2011). "New transmission Scheme for MIMO OFDM System". *International Journal of Next-Generation*, 50-62.

[11]. S. M. Alamouti. "A Simple Transmit Diversity Technique for Wireless Communications" IEEE Select. Areas in Comm., vol 16, no. 8, pp. 1451 -1458, 1998.

[12]. M. Sellathurai and S. Haykin "Space-Time Layered Information Processing for Wireless Communications" John Wiley & Sons, Inc , 1st edition, 2009.

[13]. W.C.Y. Lee "Mobile Communication Engineering: Theory & Application," 2nd Edition, McGraw-Hill International, 1998.

[14]. L. Zheng and D. Tse. Diversity and Multiplexing: A Fundamental Tradeoff in Multiple Antenna Channels. IEEE Trans. on Inform. Theory, 49(5):1073–1096, May 2003.

APPENDIX A

Thise code shows channel capacity enhancement in MIMO OFDMA system based on number of antenna.

```
clear all; clf
NR=4;NT=4; MaxIter=1000;
I=eye(NR,NR); sq2=sqrt(2); gss=['-ro';'-b^';'m-d';'-ks'];
SNRdBs=[0:2:20];
for sel_ant=1:4
for i_SNR=1:length(SNRdBs)
SNRdB = SNRdBs(i_SNR); SNR_sel_ant = 10^(SNRdB/10)/sel_ant;
%rand('seed',1);
 %randn('seed',1);
cum = 0;
for i=1:MaxIter
H = (randn(NR,NT)+j*randn(NR,NT))/sq2;
if sel_ant>NT||sel_ant<1
error('sel_ant must be between 1 and NT!');
else indices = nchoosek([1:NT],sel_ant);
end
for n=1:size(indices,1)
Hn = H(:,indices(n,:));
log_SH(n)=log2(real(det(I+SNR_sel_ant*Hn*Hn')));
end
cum = cum + max(log_SH);
end
sel_capacity(i_SNR) = cum/MaxIter;
end
plot(SNRdBs,sel_capacity,gss(sel_ant,:), 'LineWidth',2);
hold on;
end
xlabel('SNR[dB]'), ylabel('bps/Hz'),
```

```
grid on;
legend('MIMO-OFDMA NT=NR=2','MIMO-OFDMA NT=NR=3','MIMO-OFDMA NT=NR=4','MIMO-OFDMA NT=NR=5');
```

APPENDIX B

This code shows MIMO, OFDMA and MIMO OFDMA channel capacity versus SNR

```
%OFDMA channal capacity vs SNR
clear all
clc
N=2;%datastream of subchannel carrier
%rand('state' ,456321)
NR=2;
snr=0;
for i=1:10;
snr=snr+2;
for j=1:10000;
    c(j)=1/N*(NR*log(det(1+(10^snr/10)))/log(2));
end
y(i)=mean(c);
x(i)=snr;
end
figure
plot(x,y, 'bp-' , 'LineWidth' ,2);
hold on
%MIMO  Capacity  vs SNR
NR=2;%
snr=0;
for i = 1:10
snr = snr +2;
c=(NR*log(1+10^(snr/10)))/log(2);
xx(i)=snr;
yy(i)=c;
end
plot(xx,yy, 'kd-' , 'LineWidth' ,2);
%  MIMO-OFDMA capacity vs SNR
```

```
N=2;
NR=2;
%rand('state' ,456321)
snr=0;
for i=1:10;
snr=snr+2;
for j=1:10000;
    c(j)=(1/N*(NR*log(det(1+(10^snr/10)))/log(2))+(NR*log(1+10^(snr/10)))/log(2));
end
yy(i)=mean(c);
xx(i)=snr;
end
plot(xx,yy, 'mo-' , 'LineWidth' ,2)
xlabel('SNR(dB)' )
ylabel('Capacity (bit/s/Hz)' )
grid on
legend('OFDMA','MIMO' , 'MIMO-OFDMA')
title('MIMO-OFDMA Capacity' )
```

APPENDIX C

This code show on spectrum of beam forming

```
Fc=4000000;   %Carrier Frequency =1Mhz
f=Fc;
Fs=16000000;  %sampling frequency
fm=700000;
Fd=fm;
sensors=4; % number of Sensors
angle_tx=pi/4;  %Transmitter angle = 45 degrees
angle_jam=pi/4; %jammer angle = 45 degree
sens_wts=[0.2 0.4 0.6 0.8]; %Assigning Sensor weights
c=3e08;
lembda=c/f;     % Transmitted signal wavelength
samps=2*(6*Fs/fm);        %   Number of samples of Data
max=(1/Fs)*(samps-1);
t=0:1/Fs:(max);
%Modulating Data
modsignal = sin(2*pi*fm*t);           % Baseband Signal
modsignal(modsignal>0)=1;
modsignal(modsignal<0)=-1;
CAR=sin(2*pi*Fc*t); %carrier
t_sig=modsignal.*CAR;                 % modulated Data
ch_samps=length(t_sig);               %Modulated Signal Data
Samples
g=0:ch_samps-1;
figure;
d=lembda/2;  %Sensor separation
meu=15e-6;   %Step size
%FFT of Transmitted signal
k=ch_samps;
fft_samps = 2^nextpow2(k);
t_fft = fft(t_sig,fft_samps)/k;
fprime = Fs/2*linspace(0,1,fft_samps/2);
subplot(4,1,1) ;            %plot frequency components of
Transmitted Signal
plot(fprime,2*abs(t_fft(1:fft_samps/2)));
grid on;
title('Original Transmitted signal frequency spectrum');
xlabel('frequency Hz');
ylabel('magnitude');
```

```
axis([0 8e6 0 1]);
%Jammer Signal
t=0:1/Fs:max;
j_sig=1*sin(2*pi*Fc*t);
%FFT of Jamming Signal
fft_samps = 2^nextpow2(k);
j_fft = fft(j_sig,fft_samps)/k;
fprime = Fs/2*linspace(0,1,fft_samps/2);
 subplot(4,1,2);
 plot(fprime,2*abs(j_fft(1:fft_samps/2)));    %plot frequency
components  of Jamming Signal
 grid on;
 title('Spectrum of Jamming signal');
 xlabel('Frequency (Hz)');
 ylabel('magnitude');
 axis([0 8e6 0 1])
%Array Propogation Vectors
for t2=1:sensors
 v(t2)=exp(1i*(t2-1)*2*pi*d*sin(angle_tx)*1/lembda);%#ok<SAGROW>
%propagation vector
end
for t3=1:sensors
 eeta(t3)=exp(1i*(t3-
1)*2*pi*d*sin(angle_jam)*1/lembda);%#ok<SAGROW> %Propagation
Vector for Jamming Signal
end
%Jammer Reception at the Sensor Array
j_rcvd=j_sig'*eeta;    %Jamming signal Reception @ Sensors
% j_sig+t_sig = data after reception @ antenna
 x=t_sig'*v+j_rcvd;

%FFT of Recieved Signal @ Sensors
fft_samps = 2^nextpow2(k);
x_fft = fft(x,fft_samps)/k;
fprime = Fs/2*linspace(0,1,fft_samps/2);
subplot(4,1,3 );
plot(fprime,2*abs(x_fft(1:fft_samps/2))); % Plot frequency
components of Rxd signal
grid on;
title('Signal After reception at Antenna (Jamming Signal +
Desired Signal)');
```

```
xlabel('Frequency (Hz)');
ylabel('Magnitude');
axis([0 8e6 0 1])
%LMS Algorithm
 for n1=1:ch_samps
x_est=sens_wts*x';
E=t_sig-x_est;
sens_wts=sens_wts+(meu*E*x);
 end
%FFT of Estimated
fft_samps = 2^nextpow2(k);
z_fft = fft(x_est,fft_samps)/k;
fprime = Fs/2*linspace(0,1,fft_samps/2);
subplot(4,1,4);
plot(fprime,2*abs(z_fft(1:fft_samps/2))); % Plot frequency
components of Estimated signal
grid on;
title('Estimated signal frequency spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
axis([0 8e6 0 1]);
```

APPENDIX D

This code shows MIMO, OFDMA and MIMO OFDMA channel capacity versus SNR

```
%OFDMA channal capacity vs SNR
clear all
clc
N=2;%datastream of subchannel carrier
%rand('state' ,456321)
NR=2;
snr=0;
for i=1:10;
snr=snr+2;
for j=1:10000;
   c(j)=1/N*(NR*log(det(1+(10^snr/10)))/log(2));
end
y(i)=mean(c);
x(i)=snr;
end
figure
plot(x,y, 'bp-' , 'LineWidth' ,2);
hold on
%MIMO  Capacity  vs SNR
NR=2;%
snr=0;
for i = 1:10
snr = snr +2;
c=(NR*log(1+10^(snr/10)))/log(2);
xx(i)=snr;
yy(i)=c;
end
plot(xx,yy, 'kd-' , 'LineWidth' ,2);
%  MIMO-OFDMA capacity vs SNR
```

```
N=2;
NR=2;
%rand('state' ,
xx(i)=snr;
end
plot(xx,yy, 'mo-' , 'LineWidth' ,2)
xlabel('SNR(dB)' )
ylabel('Capacity (bit/s/Hz)' )
grid on
legend('OFDMA','MIMO' , 'MIMO-OFDMA' ,1)
title('MIMO-OFDMA Capacity' )
456321)
snr=0;
for i=1:10;
snr=snr+2;
for j=1:10000;
   c(j)=(1/N*(NR*log(det(1+(10^snr/10)))/log(2))+(NR*log(1+10^(snr/10)))/log(2));
end
yy(i)=mean(c);
```